

align(center + horizon)[block(width: 15cm, inset: (y: 1.1cm), stroke: (top: 2.5pt + rgb(26, 54, 93),
bottom: 0.75pt + rgb(203, 213, 225)),) [align(center)[text(size: 10.5pt, tracking: 0.35em, fill:
rgb(71, 85, 105))[大 作 业] v(0.55cm) text(size: 26pt, weight: “bold”, fill: rgb(15, 23, 42))[揭棋对弈
程序设计] v(0.65cm) text(size: 19pt, weight: “medium”, fill: rgb(30, 64, 175))[JSON 消息交互示
例] v(0.35cm) if subtitle != “” [text(size: 13pt, fill: rgb(100, 116, 139))[Message Examples –
WebSocket JSON 联调参考] v(0.35cm)]]]

v(1.5cm)

box(width: 13cm, inset: (x: 1.2cm, y: 0.95cm), fill: rgb(248, 250, 252), radius: 6pt, stroke: 0.75pt +
rgb(226, 232, 240),) [align(center)[text(size: 11pt, weight: “bold”, fill: rgb(51, 65, 85))[小组成员]
v(0.55cm) grid(columns: (1fr, 1fr), column-gutter: 1.6cm, row-gutter: 0.65cm, align: center +
horizon, [text(size: 13pt, weight: “bold”)[张恒基 (组长)]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211301 / 2024210926]], [text(size: 13pt, weight: “bold”)
[秦博宇]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211302 / 2024210940]], [text(size: 13pt, weight: “bold”)
[陈艺博]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211302 / 2024210931]], [text(size: 13pt, weight: “bold”)
[陈雨飞]
text(size: 12pt, fill: rgb(100, 116, 139))[2024211301 / 2024210918]],)]]

v(1cm) text(size: 12pt, fill: rgb(100, 116, 139))[接口文档 · 2026-06-18] v(0.5cm) text(size: 10.5pt,
fill: rgb(148, 163, 184))[北京邮电大学 · 计算机学院]]

登录

C→S Login

```
{
  "messageType": "Login",
  "userId": "player1",
  "password": "123456"
}
```

S→C loginResult

```
{
  "messageType": "loginResult",
  "success": true,
  "message": "ok",
  "userId": "player1"
}
```

匹配与准备

C→S startMatch

```
{ "messageType": "startMatch" }
```

S→C matchSuccess

```
{
  "messageType": "matchSuccess",
  "roomId": "room_a05bcbff",
  "opponentId": "player2",
  "opponentNickname": "player2"
}
```

C→S Ready

```
{ "messageType": "Ready" }
```

C→S requestFirstHand

```
{
  "messageType": "requestFirstHand",
  "wannaFirst": true
}
```

开局 gameStart

S→C (initialBoard 为 10×9 单元格数组, 此处省略为示意)

```
{
  "messageType": "gameStart",
  "redPlayerId": "player1",
  "blackPlayerId": "player2",
  "yourColor": "red",
  "firstHand": true,
  "initialBoard": [
    [{"piece": "king", "color": "black", "revealed": true}, "..."],
    "... 共 10 行 ..."
  ]
}
```

走子

C→S move

```
{  
  "messageType": "move",  
  "fromX": "b",  
  "fromY": 0,  
  "toX": "b",  
  "toY": 3,  
  "isFlip": false  
}
```

S→C moveResult (合法, 含翻子)

```
{  
  "messageType": "moveResult",  
  "success": true,  
  "valid": true,  
  "move": {  
    "from": "b0",  
    "to": "b3",  
    "type": "knight"  
  },  
  "flipResult": "knight"  
}
```

S→C moveResult (非法)

```
{  
  "messageType": "moveResult",  
  "success": true,  
  "valid": false  
}
```

S→C error

```
{  
  "messageType": "error",  
  "code": 2001,  
  "message": "非法走法"  
}
```

聊天与心跳

C→S chat

```
{  
  "messageType": "chat",  
  "content": "你好",  
  "timestamp": 1712345678901  
}
```

C→S ping / S→C pong

```
{ "messageType": "ping", "timestamp": 1712345678901 }  
{ "messageType": "pong", "timestamp": 1712345678901 }
```

提和与认输

C→S drawOffer

```
{ "messageType": "drawOffer" }
```

C→S Resign

```
{ "messageType": "Resign" }
```

终局 **gameOver**

```
{
  "messageType": "gameOver",
  "winner": "red",
  "reason": "checkmate",
  "winnerId": "player1",
  "capturedReveal": [
    { "color": "black", "wasDark": true, "piece": "rook" }
  ]
}
```

本组扩展：复盘

C→S replayRequest

```
{
  "messageType": "replayRequest",
  "stepIndex": 0
}
```

S→C replayFrame

```
{
  "messageType": "replayFrame",
  "roomId": "room_a05bcbff",
  "stepIndex": 3,
  "totalSteps": 48,
  "currentTurn": "black",
  "status": "PLAYING",
  "board": [ "... " ],
  "move": { "from": "a6", "to": "a5" }
}
```

本组扩展：再来一局 / 人机

C→S rematchRequest

```
{ "messageType": "rematchRequest" }
```

C→S startAiGame

```
{
  "messageType": "startAiGame",
  "aiLevel": "hard"
}
```

消息归属速查

messageType	归属
Login, startMatch, Ready, move, gameStart, moveResult, gameOver, chat	课程公共接口
replayRequest, replayFrame, rematchRequest, addTime, pauseGame	本组扩展

完整表见 `INTERFACE.typ` §本组扩展消息。